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float area_circle(float radius) // to calculate the area of a circle

float area_triangle(float half,floatbase,float height) // to calculate the area of a triangle

float area_rectangle(float length , float breadth) // to calculate the area of a rectangle

This can be rewritten using a single function header in the following manner

float area (float radius);

float area (float half, float base, float height);

float area (float length , float breadth);

Illustration 15.1 C++ Program to demonstrate function overloading

```
#include <iostream>
using namespace std;
void print(int i)
 {cout<< " It is integer " << i <<endl;}
void print(double  f)
{  cout<< " It is float " << f <<endl;}
void print(string c)
{  cout<< " It is string " << c <<endl;}
int main() {
        print(10);
        print(10.10);
        print("Ten");
        return 0;
}
```

Output:

```
It is integer 10
 It is float 10.1
It is string Ten
```

Tip Notes

Function overloading is not only implementing polymorphism but also reduces the number of comparisons in a program and makes the program to execute faster. It also helps the programmer by reducing the number of function names to be remembered.